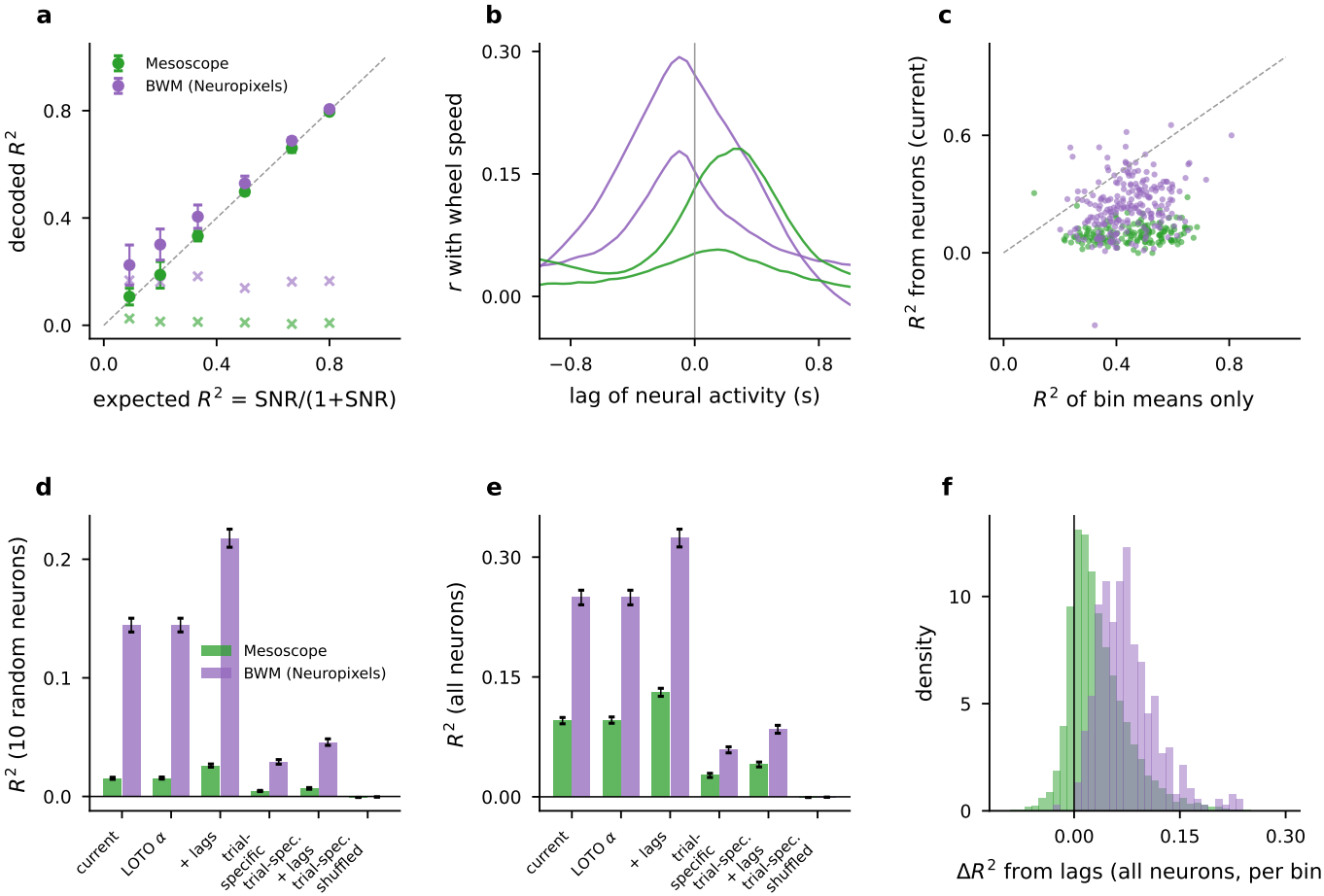

Figure 1: Checks of the wheel-speed decoding. The decoder and target are correct (after removing encoder glitches in 16 mesoscope sessions), mesoscope activity lags the wheel by about one 0.2 s bin more than spikes, and the R^2 is dominated by the stereotyped speed profile after movement onset rather than by trial-to-trial differences in speed. **a** Decoder check: a synthetic neuron (wheel speed plus Gaussian noise at signal-to-noise ratios 0.1–4) replacing one of 10 random neurons of a session is recovered at the expected R^2 (dots: mean ± s.d. over 5 draws and 3 sessions per modality; crosses: the same 10 neurons without it). The decoder reproduces the dorsal-maps GPU decoder (max difference 2×10^{-15}) and the scikit-learn reference (1×10^{-7}). The wheel-speed target agrees with brainbox's filtered speed ($r \geq 0.9999$) once encoder glitches are removed. **b** Timing at full resolution: correlation of neural activity with wheel speed at lags -1 to 1 s (positive: neural activity follows the wheel), mean over the 5 % most wheel-correlated neurons selected on the first half of each session and measured on the second half; 2 sessions per modality. Mesoscope deconvolved activity peaks 0.15–0.25 s after wheel speed, BWM spikes 0.10 s before it. **c** What the R^2 measures: predicting each bin from its mean over the training trials (no neural data) gives R^2 0.47 (mesoscope sessions, median) and 0.44 (BWM), far above decoding from all neurons of a bin (current decoder; one dot per session). Most of the variance is the stereotyped time course after movement onset; trial-to-trial differences are 53% (mesoscope) and 56% (BWM) of the variance. **d** Decoder variants, 10 random neurons per 0.4 mm bin (mean ± s.e.m. over sessions): current (leave-one-row-out choice of the ridge penalty), leave-one-trial-out (LOTO) choice, neural activity of the previous, same and next bin as features (lags), trial-specific speed (bin means of the training trials removed) without and with lags, and the trial-specific target with trials shuffled (null). Mesoscope 0.015 → 0.026 with lags, BWM 0.144 → 0.218. **e** Same from all neurons of a bin: mesoscope 0.095 → 0.131, BWM 0.249 → 0.324; trial-specific speed 0.040 vs 0.085. **f** Gain from the lagged features per (session, bin), all neurons.